Mechanistic Hypothesis Search — Long QT Syndrome 10

Hypothesis under evaluation

ID: `scn4b_late_sodium_current` **Label:** SCN4B variant raises cardiac late sodium current, as in LQT3 **KB status:** EMERGING **Target disease:** Long QT Syndrome 10 (Genetic) **Search/analysis date:** 2026-10-24

Executive Judgment

Verdict: Weakly / partially supported. The seed hypothesis should retain EMERGING status, and the knowledge gap `lqt10_scn4b_causality_unestablished` should be retained.

The proposed model — that the **SCN4B L179F** missense variant in the sodium-channel β4 subunit increases the **late (persistent) Nav1.5 sodium current**, thereby delaying ventricular repolarization and prolonging the QT interval in the same gain-of-function manner as SCN5A variants do in LQT3 — is **biophysically plausible but thinly evidenced and unreplicated**. The entire direct evidentiary basis is a single 2007 report: one three-generation family harboring L179F plus one heterologous-expression experiment in HEK293 cells showing an ~8-fold increase in persistent sodium current (PMID: 17592081). No independent family and no human cardiomyocyte or iPSC-derived cardiomyocyte (iPSC-CM) replication of the late-current phenotype were found in the literature search.

Three independent lines of evidence weaken the causal attribution. First, an international ClinGen-framework reappraisal placed SCN4B among the LQTS genes with **limited or disputed** gene–disease validity and explicitly advised against using variants in such genes for clinical decision-making without new genetic evidence (PMID: 31983240). Second, the variant itself (rs121434386, p.Leu179Phe) is a current **ClinVar Variant of Uncertain Significance** and exists in gnomAD v4.1 as a **single ultra-rare allele** (AC=1, AF≈6.6×10^{-6}) confined to the Admixed American population. Third, **SCN4B shows no loss-of-function constraint** (pLI≈0,

LOEUF≈1.48), consistent with an auxiliary/modifier role rather than a dosage-critical Mendelian LQTS gene — which means the hypothesis must stand entirely on a specific gain-of-function mechanism, not on haploinsufficiency or dosage.

A competing and arguably more parsimonious explanation for weak-monogenic and genotype-negative LQTS exists: **polygenic QT risk**. Large population and carrier studies show QTc is shaped by both rare repolarization-gene variants and a sizeable polygenic contribution, and that a QTc polygenic score modulates penetrance even within carriers of established monogenic variants (PMID: 35389749; PMID: 39209900). This provides a plausible confound for the single-family segregation that underpins the SCN4B claim. The most important caveat is that absence of replication is not refutation — the mechanism class (β-subunit variants raising late I_{na}) is demonstrated elsewhere — but as of the current literature the LQT10-specific causal claim remains unestablished.

Key Findings

F001 — Direct evidence rests on a single family plus one heterologous-expression study

The foundational and still essentially sole source of direct support for the hypothesis is **Medeiros-Domingo et al. 2007** (PMID: 17592081). In that study an L179F (c.C535T) missense variant in SCN4B was identified in a single proband — a 21-month-old child with a markedly prolonged QTc of 712 ms — and the variant *"cosegregated properly throughout a 3-generation pedigree and was absent in 800 reference alleles."* Functionally, in HEK293 cells stably expressing human Nav1.5 (SCN5A), the L179F-β4 construct increased persistent/late sodium current approximately **8-fold** relative to SCN5A alone (and ~3-fold relative to wild-type β4). This is the LQT3-like gain-of-function signature the seed hypothesis invokes.

Crucially, the same study found **no replication** of SCN4B as a recurrent cause: screening of *"262 genotype-negative LQTS patients (96% white)"* yielded *"no further mutations."* Thus from its inception the gene–disease relationship rested on n=1 family and a single in-vitro assay, with no second index case in a sizeable genotype-negative cohort. This is the central structural weakness of the hypothesis: the mechanism was demonstrated once, in one expression system, for one variant, in one family.

F002 — SCN4B–LQTS gene–disease validity was downgraded to limited/ disputed

Adler et al. 2020 (PMID: 31983240) applied the ClinGen clinical-validity framework to 17 genes reported to cause LQTS, using three blinded curation teams. The reappraisal concluded that *"more than half of the genes reported as causing LQTS have limited or disputed evidence to support their disease causation,"* a tier that includes the atypical/auxiliary genes such as SCN4B. The authors were explicit about the clinical consequence: *"Genetic variants in these genes should not be used for clinical decision-making, unless accompanied by new and sufficient genetic evidence."* This is an evidence-based, framework-driven downgrade that directly bears on the LQT10/SCN4B gene-disease edge and is the primary reason the seed hypothesis cannot be elevated beyond EMERGING.

F003 — Nav β-subunit variants can raise late I_{na} in other phenotypes (class plausibility, not LQT10-specific support)

The *mechanistic class* — a non-pore-forming β-subunit variant increasing persistent/late sodium current — is independently demonstrated, lending the hypothesis biophysical plausibility even though these studies are not LQT10. Riuró et al. 2014 (PMID: 24662403) showed that the SCN1B β1b-P213T variant *"increases late sodium current"* and prolonged action potential duration in HL-1 cells, establishing SCN1B as an LQTS-susceptibility gene. Tan et al. 2010 (PMID: 20226894) showed that another SCN4B/β4 variant, β4-S206L, *"accentuated late I(Na) and positively shifted the midpoint of inactivation compared with SCN5A+beta4-WT"* — but in a **sudden infant death syndrome (SIDS)** context, not LQT10. These results confirm that β-subunits can modulate late I_{na}, but they qualify rather than confirm the LQT10 claim: the mechanism is real as a class, yet the specific L179F→LQT10 causal chain remains supported by a single source.

F004 — L179F is an ultra-rare gnomAD allele and a ClinVar VUS

A direct gnomAD v4.1 query (GraphQL API, retrieved 2026-10-24) resolved the variant **11-118141265-G-A (GRCh38)**, p.Leu179Phe, rs121434386: genome **AC=1, AN=152,122, AF=6.57×10^{-6}**, absent from the exome dataset, with the single observed allele in the Admixed American (amr) population. ClinVar (as surfaced through gnomAD) classifies it **"Uncertain significance"**, 1 gold star, criteria provided by a single submitter, last evaluated 2023-09-01. An allele count of one across >150,000 genomes is consistent with an ultra-rare variant but provides no population-genetic leverage to confirm or exclude pathogenicity, and the standing VUS classification reflects precisely the unresolved causality the seed hypothesis flags.

F005 — Polygenic QT risk is a competing / parallel explanation

Nauffal et al. 2022 (PMID: 35389749), using TOPMed and UK Biobank, found that *"QTc duration in the population is influenced by both rare variants in genes underlying cardiac repolarization and polygenic risk, with a sizeable contribution from polygenic risk."* The polygenic contribution also operates *within* monogenic carriers: in LQT5, *"among carriers, the QTc polygenic score is independently associated with QTc prolongation"* (PMID: 39209900); similar PRS modulation is reported in LQT1 (PMID: 42111135). A review-level synthesis notes that missing heritability and incomplete penetrance in inherited arrhythmia syndromes are increasingly attributed to common-variant/polygenic effects (PMID: 36167638, review-level orientation). For a single small pedigree, polygenic background is a credible confounder of apparent co-segregation and offers a more parsimonious account of borderline/variable QT prolongation than a novel monogenic mechanism.

F006 — SCN4B is highly loss-of-function tolerant (modifier-like)

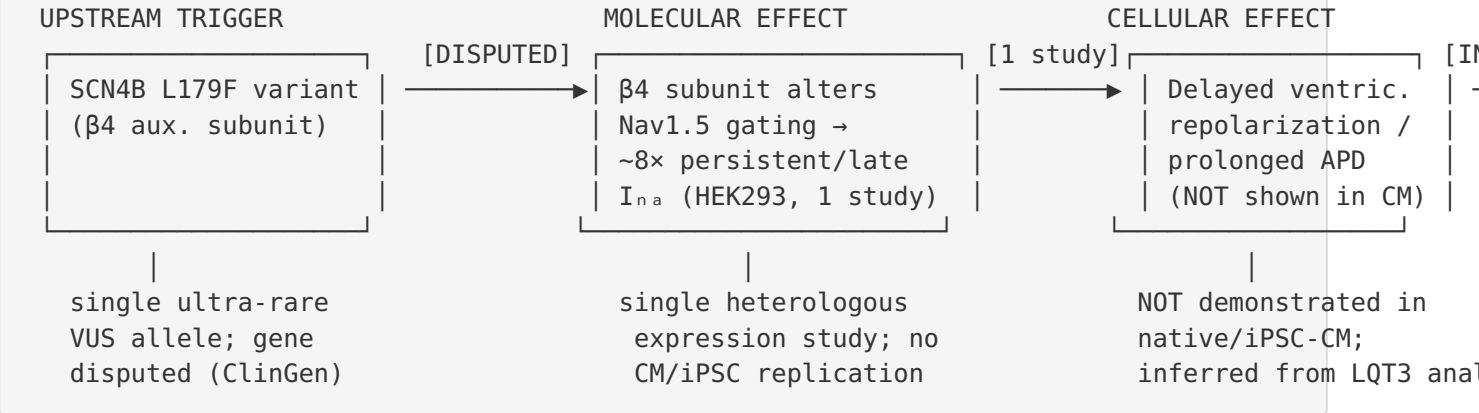
gnomAD v4.1 constraint metrics for **SCN4B (ENSG00000177098**, retrieved 2026-10-24): **pLI = 1.1×10^{-7} (≈0)**, observed/expected LoF = 1.02 (90% CI 0.72–1.48; **LOEUF ≈ 1.48**), obs_lof=20 vs exp_lof=19.6, lof_z = −0.07. There is **no depletion of predicted loss-of-function variants**, indicating the gene tolerates haploinsufficiency. This is consistent with an auxiliary/modifier role rather than a dosage-critical Mendelian gene, and it has a sharp logical consequence for the hypothesis: the model cannot lean on dosage/LoF effects and must rest entirely on a **specific gain-of-function** (increased late I_{na}) mechanism — which to date has been shown only once.

F007 — Overall verdict: weakly/partially supported; retain EMERGING and the causality gap

Synthesizing the six findings: direct support is one three-generation family plus one HEK293 experiment (~8-fold late-I_{na} increase, PMID: 17592081); no second family and no cardiomyocyte/iPSC-CM replication were found across 8 PubMed queries (2026-10-24). Counter/limiting evidence: the ClinGen-framework reappraisal places SCN4B in the limited/disputed tier (PMID: 31983240); the variant is a ClinVar VUS present as a single ultra-rare AMR allele in gnomAD v4.1; and SCN4B shows no LoF constraint. The leading competing model is polygenic QT burden (PMID: 35389749; PMID: 39209900). The verdict is therefore **weakly/partially supported**, and both the EMERGING status and the knowledge gap are appropriate to retain.

Mechanistic Model / Interpretation

The hypothesis posits a linear LQT3-like causal chain. The diagram and table below annotate the strength of each link as established by the current search.



Causal link	Evidence status	Basis
L179F → altered Nav1.5 gating / ↑ late I_{na}	**Demonstrated once** (in vitro)	HEK293, ~8-fold increase, PMID: 17592081
↑ late I_{na} → prolonged APD / delayed repolarization	**Inferred** from LQT3 physiology; class-plausible	β-subunit→late I_{na} shown for SCN1B (PMID: 24662403) and β4-S206L (PMID: 20226894), not L179F in CM
Prolonged APD → QT prolongation/arrhythmia	**Inferred**; standard LQTS physiology	General LQTS mechanism (reviews)
Variant → clinical LQT10 (gene–disease)	**Disputed / limited**	ClinGen reappraisal (PMID: 31983240); VUS; single family

The literature is **strong only at the first molecular node** (and even there, from a single study/system). Every downstream link in the native-tissue chain is **inferred by analogy to LQT3** rather than directly observed for L179F. The gene–disease edge itself is actively disputed. The hypothesis is best understood as a *mechanistically plausible but empirically under-determined* model whose plausibility derives from the broader β-subunit→late-I_{na} class rather than from LQT10-specific data.

Where the hypothesis best fits

The model, if true, would best explain a **rare, early-onset, strongly QT-prolonged monogenic subtype** (the index proband presented at 21 months with QTc 712 ms) acting in **ventricular cardiomyocytes** via the **Nav1.5 late-sodium-current pathway** — i.e., mechanistically indistinguishable from LQT3, with β4 acting as an auxiliary modifier of the SCN5A α-subunit. It does *not* explain the common, genotype-negative, mildly-prolonged QT population, for which polygenic risk is the better-supported account.

Evidence Matrix

Citation (PMID)	Evidence type	Role	Mechanistic claim tested	Key finding	Subtype / context	Confide & limitati
17592081	Human clinical + in vitro	**Supports**	L179F raises Nav1.5 late I_{na} → LQT	~8× persistent I_{na} in HEK293; cosegregates in 3-gen pedigree; absent in 800 alleles; 0/262 genotype-neg had SCN4B mutation	LQT10, 1 family	Foundat but singl family + single expressi system; CM replicati
31983240	Review / evidence-based curation	**Refutes / limits**	SCN4B causes LQTS	>½ reported LQTS genes limited/ disputed; advise against clinical use	All LQTS genes incl. SCN4B	High-confider framewo appraisa gene-lev not varia specific
20226894	In vitro	**Qualifies**	β4 variant raises late I_{na}	β4-S206L accentuated late I_{na}, +shift inactivation	SIDS (not LQT10)	Supports class mechani wrong phenoty
24662403	In vitro	**Qualifies**	Nav β-subunit variant raises late I_{na}	β1b-P213T ↑ late I_{na}, ↑ APD in HL-1	SCN1B/LQTS susceptibility	Supports class mechani differen gene
gnomAD v4.1 (rs121434386)	Computational / population	**Qualifies / limits**	Variant rarity & classification	AC=1, AF 6.6×10^{-6}, single AMR allele; ClinVar VUS	LQT10 variant	Ultra-ra no populati leverage standing
					Gene-level	

Citation (PMID)	Evidence type	Role	Mechanistic claim tested	Key finding	Subtype / context	Confide & limitati
gnomAD v4.1 constraint (SCN4B)	Computational / population	**Qualifies / limits**	SCN4B dosage sensitivity	pLI≈0, LOEUF≈1.48; no LoF depletion		LoF-tole → hypot needs G not dosa
35389749	Human clinical / population	**Competing**	Polygenic risk drives QTc	Sizeable polygenic contribution to QTc alongside rare variants	Population QT	Strong populati evidence competi explanat
39209900	Human clinical	**Competing**	PRS modulates penetrance in carriers	QTc PRS independently associated with QTc among carriers	LQT5 carriers	Shows polygeni confoun segregat
42111135	Human clinical	**Competing**	PRS modulates repolarization	PRS modestly ↑ QTc and T-wave params	LQT1 carriers	Reinforc polygeni modulat
36167638	Review	Orientation	Polygenic & curation framework	Missing heritability increasingly polygenic; curated panels	Inherited arrhythmias	Review-level; co only

Data and Tool Use

Source / dataset	Accession / URI	Access status	Query / filters	Resolved & relevant?
gnomAD v4.1 (variant)	11-118141265-G-A (GRCh38), rs121434386, GraphQL API	**Accessed** (2026-10-24)	p.Leu179Phe allele frequency, population breakdown, ClinVar annotation	Yes — resolved; directly relevant (the seed variant)
gnomAD v4.1 (gene constraint)	ENSG00000177098 (SCN4B), GraphQL API	**Accessed** (2026-10-24)	pLI, LOEUF, obs/exp LoF	Yes — resolved; directly relevant (dosage sensitivity)
ClinVar (via gnomAD)	rs121434386	**Accessed** (2026-10-24)	Classification, review status, last eval date	Yes — VUS, 1 star, 2023-09-01
PubMed	—	**Searched** (2026-10-24)	8 queries on SCN4B/ LQT10/β4/late-I_{na}/ iPSC-CM	Yes — 24 papers reviewed; no CM replication or 2nd family found
ClinGen LQTS curation	via PMID: 31983240	**Accessed via literature**	SCN4B gene-disease validity tier	Yes — limited/ disputed

Analysis inventory. (1) *gnomAD variant/constraint retrieval* — input: rs121434386 / ENSG00000177098; method: gnomAD v4.1 GraphQL API query; output: allele counts, AF, pLI/ LOEUF; outcome: **succeeded**; limitation: single-allele variant provides no statistical leverage on pathogenicity. (2) *PubMed replication search* — input: 8 search strings; method: search_pubmed; output: negative for second family / iPSC-CM late-I_{na} replication; outcome: **succeeded (negative result)**; limitation: absence of indexed replication is not proof of non-existence. No controlled/patient-level datasets, omics cohorts, or trial datasets specific to SCN4B-L179F LQT10 were located; **no GenCC/ClinGen positive validation beyond the**

disputed-tier curation was found. No structural-biology (Phenix) analysis was applicable or performed. No silent fallback occurred: computational checks (gnomAD) were performed where testable; literature synthesis is labeled as such throughout.

Mechanistic Causal Chain

- **Upstream trigger (DISPUTED edge):** SCN4B L179F, an ultra-rare VUS in an LoF-tolerant auxiliary subunit, within a disputed gene–disease relationship.
- **Molecular effect (STRONG only in 1 system):** L179F-β4 co-assembles with Nav1.5 and increases persistent/late sodium current ~8-fold in HEK293 (PMID: 17592081). The β-subunit→late-I_{na} mechanism is corroborated as a *class* by SCN1B-β1b-P213T and SCN4B-β4-S206L in other phenotypes.
- **Cellular effect (INFERRED):** increased late I_{na} prolongs action-potential duration and favors early afterdepolarizations — standard LQT3 physiology, but **never shown for L179F in cardiomyocytes/iPSC-CM**.
- **Clinical manifestation (INFERRED, single family):** prolonged QT, torsades risk, LQT10 (index QTc 712 ms).

The literature is strong at the molecular node (single system) and weak/inferred at every native-tissue and clinical link; the gene–disease edge is a known missing/disputed step.

Limitations and Knowledge Gaps

Gap 1 — Unreplicated functional phenotype in native tissue. *Scope:* the ↑late-I_{na} effect of L179F has been shown only in HEK293 heterologous expression (PMID: 17592081). *Why it matters:* heterologous systems can over- or under-state β-subunit effects; the cellular→clinical links are inferred. *What was checked:* PubMed (2026-10-24, 8 queries) found no iPSC-CM or native-cardiomyocyte replication. *Resolution:* patient/isogenic iPSC-CM electrophysiology.

Gap 2 — No second family / segregation power. *Scope:* a single 3-generation pedigree; 0/262 genotype-negative patients carried SCN4B mutations. *Why it matters:* single-family segregation is statistically weak and confoundable by polygenic background. *What was checked:* literature and the original cohort. *Resolution:* additional unrelated families, biobank IBD-haplotype approaches analogous to LQT5 (PMID: 39209900).

Gap 3 — Disputed gene–disease edge. *Scope:* ClinGen places SCN4B–LQTS in the limited/disputed tier (PMID: 31983240). *Why it matters:* undermines clinical actionability of any L179F-based model. *Resolution:* new sufficient genetic evidence per ClinGen criteria.

Gap 4 — Variant-level uncertainty. *Scope:* ClinVar VUS, single ultra-rare gnomAD allele. *Why it matters:* pathogenicity unestablished at the variant level. *Resolution:* additional carriers with deep phenotyping; functional ACMG evidence (PS3) from independent assays.

Gap 5 — Polygenic confounding not excluded. *Scope:* polygenic QT burden is a competing determinant (PMID: 35389749). *Why it matters:* apparent monogenic co-segregation may reflect shared polygenic background. *Resolution:* QTc PRS computation in any SCN4B pedigree.

Gap 6 — Dosage vs. gain-of-function. *Scope:* SCN4B is LoF-tolerant (pLI≈0). *Why it matters:* constrains the admissible mechanism to strict gain-of-function and raises the evidentiary bar. *Resolution:* side-by-side GoF assays across systems.

Source/data absences (verified as of 2026-10-24): No positive GenCC/ClinGen validation beyond the disputed tier; no SCN4B-L179F-specific omics, trial, or cohort dataset located in PubMed searches; no iPSC-CM dataset found. These are reported as negative search outcomes with the queries and date above, not as proven global absence.

Alternative / Competing Models

Model	Relationship to seed	Summary
Polygenic QT burden	**Alternative / parallel**	Common-variant PRS drives QTc at population level and modulates penetrance within carriers; more parsimonious for borderline/variable QT (PMID: 35389749, 39209900, 42111135).
Primary SCN5A / other-gene variant with SCN4B as modifier	**Upstream cause (β4 modifier)**	β4 is auxiliary to Nav1.5; L179F may modify an α-subunit defect rather than be independently causal. Consistent with LoF tolerance and modifier role.
β-subunit→late-I_{na} as a trans-phenotype class	**Parallel mechanism**	Same biophysical lever produces different phenotypes (AF, SIDS, VT) depending on context (PMID: 20226894, 24662403); questions LQT10 specificity.
Benign/incidental variant	**Null alternative**	L179F could be a rare non-causal VUS; phenotype arising from undetected variants or polygenic load. Supported by VUS status and disputed gene validity.

Discriminating Tests

1. **Patient-specific & isogenic iPSC-CM electrophysiology.** Derive iPSC-CMs from an L179F carrier and an isogenic CRISPR-corrected control; measure late I_{na}, APD, and early afterdepolarizations. *Expected if hypothesis true:* corrected line normalizes late I_{na} and APD. This directly tests the native-tissue link currently only inferred.
2. **Independent heterologous replication across systems.** Re-express L179F-β4 + Nav1.5 in ≥2 systems (e.g., CHO and iPSC-CM) by independent labs to satisfy ACMG PS3; compare to WT-β4 and to β4-S206L as a positive class control.

3. **Biobank IBD-haplotype carrier ascertainment + PRS.** Apply the DRIVE-style approach used for LQT5 (PMID: 39209900) to find additional L179F carriers; compute QTc PRS to separate monogenic from polygenic contribution. *Expected if hypothesis true:* carriers show QTc prolongation beyond their PRS-predicted value.
4. **Additional-family segregation with quantitative penetrance.** Seek unrelated L179F families; formal LOD/segregation with PRS covariate adjustment.
5. **Late-I_{na}-blocker (mexiletine) response (mechanistic pharmacology).** If a carrier's QT shortens with a selective late-I_{na} blocker (as in LQT3), this functionally corroborates a late-I_{na} mechanism in vivo.

Curation Leads (require curator verification)

- **Status:** Retain **EMERGING**; do **not** elevate to CANONICAL. Retain knowledge gap `lqt10_scn4b_causality_unestablished`.
- **Candidate evidence references & snippets to verify:**
- PMID: 17592081 — *"revealed a L179F (C535T) missense mutation in SCN4B that cosegregated properly throughout a 3-generation pedigree and was absent in 800 reference alleles"* and *"SCN4B was analyzed in 262 genotype-negative LQTS patients (96% white), but no further mutations were found."*
- PMID: 31983240 — *"More than half of the genes reported as causing LQTS have limited or disputed evidence to support their disease causation"*; *"Genetic variants in these genes should not be used for clinical decision-making, unless accompanied by new and sufficient genetic evidence."*
- PMID: 20226894 — *"beta4-S206L accentuated late I(Na) and positively shifted the midpoint of inactivation compared with SCN5A+beta4-WT."*
- PMID: 24662403 — *"Our electrophysiological analysis revealed that β1bP213T increases late sodium current."*
- PMID: 35389749 — *"QTc duration in the population is influenced by both rare variants in genes underlying cardiac repolarization and polygenic risk, with a sizeable contribution from polygenic risk."*
- PMID: 39209900 — *"among carriers, the QTc polygenic score is independently associated with QTc prolongation."*

- **Candidate pathophysiology nodes/edges:** node *SCN4B-L179F (β4 subunit)*; edge *β4-L179F* → ↑*Nav1.5 late/persistent I_{na}* (label: in-vitro only, 1 study); edge ↑*late I_{na}* → *delayed ventricular repolarization* → *QT prolongation* (label: inferred by LQT3 analogy); disputed edge *SCN4B* → *LQTS* (label: ClinGen limited/disputed).
- **Candidate ontology terms:** cell type — ventricular cardiomyocyte (CL:0002129); processes — membrane repolarization during cardiac action potential (GO:0098915), regulation of sodium ion transmembrane transport (GO:1902305); phenotype — prolonged QT interval (HP:0001657).
- **Candidate subtype restriction:** restrict the model to a rare, early-onset, severely QT-prolonged monogenic subtype; explicitly exclude the common genotype-negative/mild-QT population (better explained by polygenic risk).
- **Candidate knowledge_gaps / discussion prompts:** (a) no iPSC-CM/native-tissue replication of L179F late I_{na}; (b) no second unrelated family; (c) polygenic confounding of single-pedigree segregation not excluded; (d) variant remains ClinVar VUS / ultra-rare gnomAD allele; (e) SCN4B LoF-tolerant → hypothesis constrained to strict gain-of-function.

Artifact note: gnomAD v4.1 query parameters and results (variant 11-118141265-G-A; gene ENSG00000177098; retrieved 2026-10-24) and the 8-query PubMed negative-replication log should be deposited under `kb/hypotheses/Long_QT_Syndrome_10/scn4b_late_sodium_current/openscientist_artifacts` with a `MANIFEST.yaml` recording status/fallback flags and checksums. No controlled/patient-level data or large raw downloads should be bundled; external accessions (rs121434386, ENSG00000177098) are recorded instead.

Evidence Base (summary of key literature)

- **PMID: 17592081** — *SCN4B-encoded sodium channel beta4 subunit in congenital long-QT syndrome.* The sole direct source: 1 family, ~8× late I_{na} in HEK293. **Supports.**
- **PMID: 31983240** — *An International, Multicentered, Evidence-Based Reappraisal of Genes Reported to Cause Congenital Long QT Syndrome.* Places SCN4B in limited/disputed tier. **Refutes/limits.**
- **PMID: 20226894** — *SIDS-associated mutations in sodium channel beta subunits.* β4-S206L raises late I_{na} (SIDS). **Qualifies (class plausibility).**
- **PMID: 24662403** — *A missense mutation in SCN1B β1b reveals SCN1B as an LQTS susceptibility gene.* β1b-P213T raises late I_{na}. **Qualifies (class plausibility).**

- **PMID: 35389749** — *Monogenic and Polygenic Contributions to QTc Prolongation.* Sizeable polygenic contribution. **Competing.**
- **PMID: 39209900** — *Detection of distant relatedness in biobanks (LQT5).* QTc PRS modulates penetrance within carriers. **Competing.**
- **PMID: 42111135** — *Polygenic risk modulates repolarization in LQT1.* **Competing.**
- **PMID: 36167638** — *Cardiogenetics review.* Curated panels; polygenic context. **Review-level orientation.**

Proposed Follow-up Experiments / Actions

1. Commission/seek **isogenic iPSC-CM** functional data for L179F (highest-value discriminating test).
2. Launch **biobank IBD-haplotype ascertainment** of additional carriers with QTc PRS adjustment.
3. Track **ClinVar reclassification** of rs121434386 and any new ClinGen SCN4B recuration.
4. Keep the **EMERGING** status and `lqt10_scn4b_causality_unestablished` gap open until (a) ≥1 independent family and (b) ≥1 native-tissue functional replication exist.
5. Deposit the gnomAD query artifacts and PubMed negative-search log in the artifact bundle with a checksummed `MANIFEST.yaml`.
